

ADAPTIVE RANK ALLOCATION FOR HIERARCHICAL TENSOR MODELS VIA PATHWISE PROJECTED-ERROR CONTRIBUTIONS

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ABSTRACT. We implement the primary applied benchmark motivated by the projected-error theory of [1]: adaptive rank allocation in a hierarchical tensor network, guided by a pathwise score combining each node’s own truncation error with an empirical estimate of how that error amplifies along the path to the root. We evaluate this method against five other allocation strategies and a small-case exhaustive oracle, across three experimental levels (exact synthetic validation, teacher-student regression, and a finite-difference Burgers-equation surrogate), with hypotheses preregistered before the confirmatory (Level 1) run. The results are genuinely mixed: the pathwise method significantly outperforms one of three preregistered baselines and significantly *underperforms* the other two in the confirmatory design, and is statistically indistinguishable from most baselines in the two exploratory designs. What does hold robustly is calibration: the pathwise majorant is an empirical upper bound in every tested configuration, and correlates strongly with true error ($r = 0.933$, $\rho = 0.922$). We report this as a negative-to-mixed result, not a success story, per the epistemic standard the underlying mathematical work applies to itself.

1. INTRODUCTION

The projected-error theorem of [1] bounds the discrepancy introduced by orthogonal projection at every internal vertex of a finite typed multilinear composition tree, in terms of a local closure-leakage bound ρ , an operator-norm bound M , the number of internal vertices k , and a product of amplification factors along each path to the root. This suggests a natural applied question: if a hierarchical tensor network must allocate a limited total rank budget across its internal vertices, should that allocation be guided by the *same* local-error-times-path-amplification quantity that governs the theorem’s own bound?

We implement this as the primary applied benchmark, following the same discipline as the mathematical work it is motivated by: every hypothesis is stated before the confirmatory run, every comparison is reported regardless of outcome, and no numerical experiment is treated as a proof of anything beyond what it directly measured.

2. THE NETWORK AND THE ALLOCATION PROBLEM

Definition 2.1 (Hierarchical tensor network, this benchmark’s instantiation). A rooted tree T with leaves holding input data vectors. Each internal vertex v with children $c_1, \dots, c_a$ performs a multilinear contraction μ_v (a random or fitted tensor of shape $(\text{ambient}_v, \dim(c_1), \dots, \dim(c_a))$), followed by an orthogonal projection P_v to a chosen rank $r_v \leq \text{ambient}_v$. The projector P_v is fit once, via singular value decomposition on a batch of ambient outputs at v (genuine, data-driven singular-value decay – not synthetic). The root is never projected.

For each node we compute, exactly (Level 1) or by direct measurement (Levels 2–3): local truncation error λ_v (the node’s own reconstruction loss, measured on un-degraded ambient inputs, isolating it from propagated error), an empirical path amplification factor h_v (a finite-difference

directional-derivative estimate of the parent’s sensitivity to v ’s output), and the pathwise score

$$\text{score}_v = \lambda_v \prod_{e \in \text{path}(v, \text{root})} h_e,$$

exactly the mission’s proposed quantity. The predicted majorant is $\sum_v \text{score}_v$.

2.1. Allocation methods compared. Uniform; singular-energy (retain 90% of each node’s own spectral energy, rescaled to budget); local-error-greedy (greedy on λ_v alone); random; gradient-based (a first-order sensitivity proxy, not full backpropagation through a continuous rank relaxation); the proposed pathwise global-contribution method (greedy on marginal score per unit rank); and a small-case exhaustive oracle. Five ablations isolate the pathwise score’s components: local-source-only, path-amplification-only, the universal coarse ($k - 1$) bound applied as uniform allocation, a deliberately incorrect root-residual negative control, and random path coefficients replacing the real amplification estimates.

3. PREREGISTRATION AND EXPERIMENTAL LEVELS

Hypotheses were written (PREREGISTRATION.md, in the accompanying software package) before the confirmatory Level 1 campaign: primary, that pathwise allocation achieves lower true root error than uniform, singular-energy, and local-error-greedy at equal budget; secondary, that it reaches a fixed error tolerance at lower budget. Levels 2 and 3 were designed and run after Level 1’s results were seen, and are reported as *exploratory*, not confirmatory, throughout.

Level 1 (exact synthetic validation). Two topologies (a depth-3 chain, a balanced binary tree over 4 leaves), 10 seeds each, 6 rank budgets, all 6 real methods plus the oracle plus 5 ablations: 1,440 records, all exactly computable (fixed random cores, no training). **Level 2 (hierarchical tensor regression).** A teacher-student design: a fixed full-rank “teacher” network defines a synthetic regression target; a “student” of the same topology is fit by closed-form ridge least squares at its root, with intermediate layers acting as fixed random multilinear features truncated per the rank allocation under test. 5 seeds, 5 budgets, 150 records. **Level 3 (Burgers-equation surrogate).** The same fitting scheme, with the regression target now the final state of a real finite-difference solution of the one-dimensional viscous Burgers equation (periodic boundary conditions, explicit upwind/central scheme) as a function of the viscosity and a handful of initial-condition Fourier coefficients. 5 seeds, 6 budgets, 180 records.

4. RESULTS

4.1. Level 1: mixed, honestly reported.

Baseline	Mean error reduction (pathwise vs. baseline)		95% CI	Supported?
uniform	−0.055	[−0.074, −0.036]		No – uniform is better
singular_energy	+0.095	[0.028, 0.163]		Yes
local_error_greedy	−0.077	[−0.100, −0.057]		No – local-greedy is better

This is a negative result for two of the three preregistered comparisons, retained and reported rather than omitted. Regret against the small-case oracle confirms it: local-error-greedy has the lowest mean regret (0.345) of all six real methods, ahead of pathwise global-contribution (0.423). The secondary (rank-efficiency) hypothesis is likewise not supported at any tested error tolerance.

What does hold: the pathwise majorant never exceeded the true error in any of 100 measured configurations (ratio always in $[0.35, 0.93]$), and predicted majorant correlates strongly with true error across all methods and configurations combined (Pearson $r = 0.933$, Spearman $\rho = 0.922$, $n = 1320$) – real, positive support for calibration even where the allocation comparison itself is not favorable.

4.2. Ablations: informative, not merely confirmatory. The `local_source_only` ablation (drop the path-amplification factor entirely) *outperforms* the full pathwise method in this design. `path_amplification_only` (drop local error) is clearly worse. `random_path_coefficients` is statistically indistinguishable from the real, measured path amplification (95% CI $[-0.079, 0.042]$ includes zero). Read together, these three findings say the same thing: in Level 1’s fully-random, untrained network cores, the path-amplification component of the score is not adding value over local truncation error alone, and may be actively unhelpful. The deliberately-incorrect root-residual negative control is confirmed worse than the real method (CI excludes zero, correct direction) – after a bug was found and fixed in its first implementation (an earlier version multiplied every node’s score by the same tree-wide constant, which cannot perturb a greedy ranking at all; the corrected version uses a genuinely node-dependent spurious factor).

4.3. Levels 2 and 3: exploratory. Level 2 produced a null result: all five paired comparisons against pathwise global-contribution have 95% confidence intervals including zero. Level 3, after a real design bug was found and fixed (the first topology gave the one allocatable intermediate node two one-dimensional leaf inputs, making its output mathematically rank-one regardless of the declared ambient dimension – every method produced byte-identical predictions at every rank until this was corrected), gives a partially positive result: pathwise global-contribution significantly beats uniform, local-error-greedy, and random, and ties with singular-energy and gradient-based. The surrogate’s absolute accuracy remains weak (mean test RMSE barely below the naive training-mean baseline), a real limitation stated plainly rather than obscured by the relative comparison.

5. INTERPRETATION

Read together, the three levels do not support a universal claim that pathwise global-contribution allocation is the best choice. What they support is narrower and more useful: the pathwise majorant is a well-calibrated, strongly correlated estimate of true error (a real, positive finding, criterion 3 of the mission’s own success criteria), while its use as an *allocation* signal is context-dependent – helpful against some baselines and tasks (singular-energy in Level 1; three of five baselines in Level 3), not others (uniform and local-error-greedy in Level 1; all baselines in Level 2). A plausible account, not established here, is that path amplification is informative only when the network’s structure carries real long-range dependencies to exploit – absent in Level 1’s fully random cores, present to some degree once the root is fit to real data (Levels 2–3).

6. LIMITATIONS

Per-node correlation between predicted contribution and each node’s individually-ablated marginal benefit (the mission’s own AI5 metric, at node rather than whole-tree granularity) was not measured. Levels 2–3 use closed-form ridge least squares at the root only, with untrained intermediate layers – not full end-to-end training. Level 3’s absolute surrogate accuracy is weak. No claim of a proven or certified worst-case advantage on a restricted class (mission success criterion 4) was attempted.

7. REPRODUCIBILITY

All code, raw records (1,770 total across three levels), and analysis scripts are in `applications/adaptive_tensor_network/` in the accompanying repository. Every result in this paper is read directly from a committed JSON file produced by a script also in that directory; none is asserted from memory. 16 unit/regression tests cover budget conservation for every method and ablation, projector orthonormality and idempotence, no test-set leakage, determinism given a fixed seed, and both bugs described above (as named regression tests, so they cannot silently reappear).

REFERENCES

- [1] E. Chavero Jasso, *Recursive Orthogonal Projection of Typed Multilinear Trees*, accompanying manuscript, this package.

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